AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2025

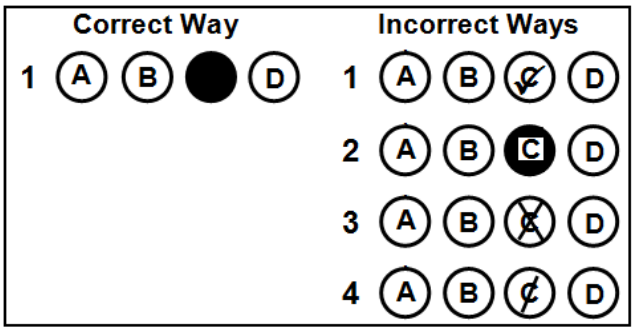
Computer Science Paper I

Time: 1 hour 30 minutes Marks: 50

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. The Operating System (OS) that permits several users at different terminals to access a specific computer system simultaneously is

   A. simple batch OS.
   B. multitasking OS.
   C. time-sharing OS.
   D. distributed OS.

2. Consider the following 4 processes with burst times:

   - Process P1: 6 ms
   - Process P2: 2 ms
   - Process P3: 8 ms
   - Process P4: 3 ms

   Which of the following sequences of execution will minimise the total execution time?

   A. P1 → P2 → P3 → P4
   B. P2 → P4 → P1 → P3
   C. P3 → P1 → P2 → P4
   D. P4 → P2 → P1 → P3

3. The advantage of using the Command Line Interface (CLI) over the Graphical User Interface (GUI) is that CLI

   A. requires more system resources.
   B. is more user-friendly for beginners.
   C. offers better multitasking capabilities with graphical windows.
   D. allows more precise control with faster execution of commands.

4. In an operating system, the function of process management is to

   A. manage the user interface.
   B. allocate system resources to processes.
   C. schedule updates for software packages.
   D. ensure communication between the network and applications.

5. The MAIN purpose of a multithreading is to

   A. allow a single CPU to run multiple operating systems.
   B. use additional memory to improve system performance.
   C. execute multiple programs in parallel on separate processors.
   D. perform multiple tasks concurrently within the same process.

6. Disk Operating System (DOS) is an example of

   A. time-sharing operating system.
   B. single-user operating system.
   C. multi-user operating system.
   D. real-time operating system.

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7. The phase that specifies the software architecture and system structure in System Development Life Cycle (SDLC) is

   A. implementation.
   B. deployment.
   C. planning.
   D. design.

8. A computer system is defined as a

   A. set of instructions that tells the computer what to do.
   B. database that stores user data and application information.
   C. program that runs on a computer to manage hardware resources.
   D. collection of interconnected components that achieve a goal together.

9. Which of the following statements defines the term, 'reserved words'?

   A. They are optional and can be chosen by the programmer.
   B. They are predefined keywords with a specific meaning.
   C. They can be used only for the documentation purpose.
   D. They can be used as variable names.

10. The CORRECT format of the main structure in C++ program is

    A. char main( )
    B. int main(void)
    C. double main( )
    D. float main(void)

11. Consider the given variable names:

    I. num_1
    II. 1_num
    III. num-1

    According to the rules of naming variables in C++, the VALID variable name(s) is/ are

    A. I only.
    B. III only.
    C. I and II.
    D. II and III.

12. A developer wants to store the age and height of a student in a C++ program.

    The VALID variable declaration and data type is

    A. int b; and char c;
    B. int a; and float b;
    C. long int b; and char c;
    D. long int a; and long float b;

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13. The CORRECT way to write a multi-line comment in C++ is

A. *\This is a comment*
B. */ This is a comment/*
C. /* This is a comment */
D. *This is a comment*\

14. Consider the following program:

```
#include<iostream>
using namespace std;
void my_function( );
int  var1 = 10;

void my_function( )
{
   int var2 = 5;
   cout << var1 << endl;
   cout <<var2 << endl;
}
int main( )
{
   int var3 = 8;
   float var4 = 45.5;
   my_function( );
   return 0;
}
```

The global variable in this program is

A. var1
B. var2
C. var3
D. var4

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15. Consider the following program:

```
#include <iostream>
using namespace std;

int main( )
{
   int a = 15, b = 4;
   int result = a / b;
   cout << "Result: " << result << endl;
   return 0;

}
```

The output of the given program is

A. Result: 3
B. Result: 3.0
C. Result: 3.000
D. Result: 3.75

16. Consider the following program:

```
#include<iostream>
using namespace std;

int main( )
{
   int a3 = 8;
   float b4 = 45.5;
   cout<< ++a3 <<"\n" << b4++ <<"\n" ;
   cout << ++b4 <<"\n" << a3++ <<"\n" ;
return 0;
}
```

The output of the given program is

8 45.5 46.5 9	9 45.5 46.5 10
A	B
9 46.5 47.5 9	9 45.5 47.5 9
C	D

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17. Consider the following program:

```
#include<iostream>
using namespace std;

int main( )
{
   int a = 4, b = 5;
   if (a <= b)
      a  = ++a * 6;
      b  = b++ - 3;
   cout << a << “and“ << b << endl;
   return 0;
}
```

The output of the given program is

A. 24 and 2
B. 25 and 3
C. 30 and 3
D. 30 and 2

18. The number of bytes an unsigned long int occupies is

A. 2
B. 4
C. 8
D. 10

19. A company is awarding a bonus to employees based on their years of services:

- Employees with more than 10 years of services will get a 20% bonus.
- Employees with 5 to 10 years of services will get a 10% bonus.
- Employees with fewer than 5 years of services will get a 5% bonus.

If the years_of_service is equal to 7, the code snippets that will implement the given logic is

if(years_of_service > 10) cout << "Bonus: 20%"; else if(years_of_service >= 5) cout << "Bonus: 5%"; else cout << "Bonus: 10%";	if(years_of_service > 10) cout << "Bonus: 10%"; else if(years_of_service < 5) cout << "Bonus: 5%"; else cout << "Bonus: 20%";
A	B
if(years_of_service > 10) cout << "Bonus: 20%"; else if(years_of_service >= 5) cout << "Bonus: 10%"; else cout << "Bonus: 5%";	if(years_of_service >= 5) cout << "Bonus: 5%"; else if(years_of_service > 10) cout << "Bonus: 20%"; else cout << "Bonus: 10%";
C	D

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20. In a program, a switch statement without a 'break' will

A. continue to execute the next case statement.
B. go directly to the default case.
C. give a syntax error.
D. stop the execution.

21. The use of exit( ) function in a C++ program is to terminate the

A. built-in function.
B. current structure.
C. user-defined function.
D. entire program at once.

22. Consider the following program:

```
#include<iostream>
using namespace std;

int main( )
{
   int a = 1;
   int n;
   cout<<"\n Enter the value of n: ";
   cin >> n;
         for(; a<=n; a = a + 1)
   cout << a << endl;
   return 0;
}
```

If the value of n is 5, then the output of the given program will be

2 3 4 5 6	1 2 3 4 5
A	B
0 0 0 0 0	1 3 5 7 9
C	D

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23. Consider the following program:

```
 #include<iostream>
using namespace std;

int main( )
{
   int k = 0, n = 5;
   while(k<=n) {
     cout << n << endl;
     ++k;
   }
   return 0;
}
```

How many times the while loop will execute in the given program?

A. 4
B. 5
C. 6
D. 7

24. Consider the following program:

```
#include<iostream>
using namespace std;

int main( )
{
   int i = 2;
   do
   {
     i = i * 3;
     ++i;
   }
   while(i<=15);
   cout << i;
   return 0;
}
```

The output of the given program is

A. 7
B. 15
C. 21
D. 22

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25. Consider the following program:

```
#include <iostream>
using namespace std;

int main( )
{
   for (int i = 0; i < 5; ++i)
    {
      if (i == 3)
          break;
      cout << i << " ";
    }
   return 0;
}
```

The output of this program is

A. 0 1
B. 0 1 2
C. 0 1 2 3
D. 0 1 2 3 4

26. All of the following objectives can be achieved using a loop statement EXCEPT

A. print first 10 numbers of fibonacci series.
B. print the sum of first 5 odd numbers.
C. determine the validity of a triangle.
D. calculate the factorial of a number.

27. An array of 5 integers in C++ program is declared as

A. int ar[5];
B. ar int[5];
C. int[5] ar;
D. int ar = new int[5];

28. The size of array in C++ program is specified inside

A. round brackets.
B. square brackets.
C. curly brackets.
D. angle brackets.

29. The function that is used to append a string onto the end of another string is

A. strcat( )
B. strlen( )
C. strcpy( )
D. strcmp( )

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30. All of the following are the disadvantages of user-defined functions EXCEPT that it

A. makes the code run slower.
B. eliminates the need for variables.
C. increases the complexity of the code.
D. allows code for reuse and modularity.

31. Consider the given function.

double fact(int a)

The word 'double' in the given function represents

A. name of the function.
B. function's arguments.
C. data type of the argument.
D. return type of the function.

32. Consider the following program:

```
#include <iostream>
using namespace std;

float mply(float a, float b)
{
   return a * b;
}

int main( )
{
   float result ;
   result = mply(4.5, 3.2);
   cout << "The product is:" << result;
   return 0;
}
```

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The output of this program is

A. 1.3
B. 1.4
C. 7.7
D. 14.4

33. Consider the following program:

```
#include <iostream>
using namespace std;
void random(int a, int b);
void random(int a, int b)
{
cout << " A is: " << a+22 << endl;
cout << " B is: " << b-5 << endl;
}
int main( )
{
  random(20, 30.50);
  return 0;
}
```

The output of this program is

A is: 22 B is: 50.50	A is: 42 B is: 25.50
A	B
A is: 42 B is: 25	A is: 22 B is: 80.0
C	D

34. Consider the given list of functions.

I. div( )
II. add( )
III. sub( )
IV. abs( )

The built-in functions in this list are

A. I and II.
B. I and IV.
C. II and III.
D. III and IV.

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Use the given program to answer Q.35 and Q.36.

```
#include <iostream>
using namespace std;
int YZ(int s, int t);
int main( )
{
  int c=10, d=5, u;
  u = YZ(c,d);
  cout<<u+2;
  return 0;
}
int YZ(int s, int t)
{
   int x;
   x=(s/t)%5;
   return x;
}
```

35. The statement that shows the declaration of user-defined function in the given program is

A. int YZ(int s, int t);
B. int YZ(int s, int t)
C. x=(s+t)%5;
D. u=YZ(c,d)

36. The number of actual arguments in the given program is

A. two.
B. three.
C. four.
D. five.

37. Consider the given part of a C++ program.

```
int a=0;
int b=&a;
int *c;
int e=5;
int d=&e;
q++;
--x;
```

How many times the reference operator is used in this program?

A. 0
B. 1
C. 2
D. 3

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38. Consider the following program:

```
#include <iostream>
using namespace std;
int main( )
{
   int ca = 2.5;
   int *cb;
   cb = &ca;
   ca = 5.5;
   cout << "ca = "<< ca <<"\n";
   cout << "cb = " << *cb;
}
```

The output of the given program is

ca: 5 cb: 5	ca: 2.5 cb: 5.5
A	B
ca: 5.5 cb: 2.5	ca: 8 cb: 7
C	D

39. The CORRECT way to declare a pointer variable is

A. int x*;
B. int *x;
C. int &x;
D. int x&;

40. In object oriented programming, a class is defined as

A. built-in data type.
B. built-in characters.
C. user-defined data type.
D. user-defined characters.

41. All of the following are the important pillars of Object-Oriented Programming (OOP) EXCEPT

A. abstraction.
B. typecasting.
C. encapsulation.
D. polymorphism.

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42. Consider the given C++ program.

```
#include <iostream>
using namespace std;
class NewCl {
private:
   int privateVal=3;

public:
   NewCl( ) {
      privateVal = 25;
   }
   int getPrivateVal( ) {
   return privateVal;
   }
};
int main( ) {
   NewCl obj;
   int value = obj.getPrivateVal( );
   cout << value << endl;
   return 0;
}
```

The output of this program is

A. 3
B. 7
C. 15
D. 25

43. Which of the following is FALSE about constructors?

A. It has integer return type.
B. It initialises objects of a class.
C. It can take one or more arguments.
D. Its name is the same as class name.

44. The term that is used for the OOP concept where method and attributes are combined together into a single entity is

A. abstraction.
B. inheritance.
C. encapsulation.
D. polymorphism.

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45. The data member and member functions can be accessed in the same class and the derived class.

With reference to the given statement, the option that shows the access specifiers used to give access to these members is

	Access Specifiers		
	Public	**Protected**	**Private**
A	Yes	Yes	Yes
B	Yes	Yes	No
C	No	No	Yes
D	No	Yes	Yes

46. Operator overloading and function overloading are the examples of

A. abstraction.
B. inheritance.
C. encapsulation.
D. polymorphism.

47. The syntax that derives a class **Z** from two base classes **S** and **T** is

A. class Z : public S, public T
B. public S, public T : class Z
C. class Z : public S & T
D. public S & T : class Z

48. In C++ program, the function that is used to open a file is

A. fopen( )
B. ifstream( )
C. ofstream( )
D. file.open( )

49. Which of the following classes are related to C++ file handling with respect to input operations?

I. ofstream
II. ifstream
III. fstream

A. I only
B. II only
C. I and III
D. II and III

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50. A sample file named as xyz.txt has the following data.

Ali 20

To read data from this file, the character array(s) required is/ are

A. 1
B. 2
C. 5
D. 6

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END OF PAPER

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